

# STT 861: Theory of Probability & Statistics I

## Lecture 11: Expected Values & Jensen's Inequality

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Casella & Berger, §2.2, pp. 55–58

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# Motivation — What Is an Expected Value?

## Intuition

The **expected value** of  $X$  is its *probability-weighted average* — the long-run average value if the experiment were repeated many times.

**Fair die example.** Roll a fair die:  $X \in \{1, 2, 3, 4, 5, 6\}$ , each with prob  $1/6$ .

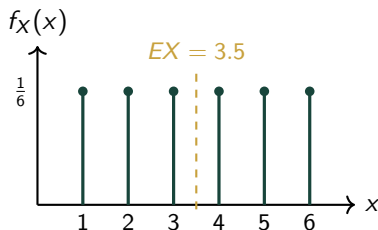
$$EX = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = 3.5$$

You never actually roll a 3.5 — but over many rolls the average converges to 3.5.

**Biased example.** Suppose  $P(X = 1) = 0.9$ ,  $P(X = 10) = 0.1$ .

$$EX = 0.9 \times 1 + 0.1 \times 10 = 0.9 + 1.0 = 1.9$$

The rare large value *pulls* the mean up.



## Definition 2.2.1 — Expected Value / Mean

### Definition 2.2.1 (CB)

The **expected value** (or **mean**) of a random variable  $g(X)$ , denoted  $E g(X)$ , is:

$$E g(X) = \begin{cases} \int_{-\infty}^{\infty} g(x) f_X(x) dx & \text{if } X \text{ is continuous} \\ \sum_{x \in \mathcal{X}} g(x) f_X(x) = \sum_{x \in \mathcal{X}} g(x) P(X = x) & \text{if } X \text{ is discrete} \end{cases}$$

provided the integral (or sum) exists. If  $E|g(X)| = \infty$ , we say  $E g(X)$  **does not exist**.

### Special cases

- $g(x) = x$ :  $EX =$  the **mean** of  $X$ , also written  $\mu$ .
- $g(x) = x^2$ :  $EX^2 =$  the **second moment**.
- $g(x) = (x - \mu)^2$ :  $E(X - \mu)^2 =$  the **variance**.
- $g(x) = c$  (constant):  $Ec = c$ .

### Two ways to compute $Eg(X)$

**Method 1** (direct): use the definition with  $f_X$ .

$$Eg(X) = \int g(x) f_X(x) dx$$

**Method 2** (transform): find  $f_Y$  where  $Y = g(X)$ , then

$$Eg(X) = EY = \int y f_Y(y) dy$$

Both give the same answer (Law of the Unconscious Statistician).

## Example 2.2.2 — Exponential Mean

### Example 2.2.2 (CB): Exponential( $\lambda$ )

$$f_X(x) = \frac{1}{\lambda} e^{-x/\lambda}, \quad x > 0, \quad \lambda > 0.$$

$$EX = \int_0^{\infty} x \cdot \frac{1}{\lambda} e^{-x/\lambda} dx$$

Integrate by parts:  $u = x$ ,  $dv = \frac{1}{\lambda} e^{-x/\lambda} dx$ :

$$\begin{aligned} EX &= \left[ -x e^{-x/\lambda} \right]_0^{\infty} + \int_0^{\infty} e^{-x/\lambda} dx \\ &= 0 + \left[ -\lambda e^{-x/\lambda} \right]_0^{\infty} = \lambda \end{aligned}$$

## Interpretation

For an  $\text{Exponential}(\lambda)$  random variable (e.g. waiting time), the mean waiting time is  $\lambda$ . The parameter  $\lambda$  serves double duty: it appears in both the pdf *and* equals the mean. (Some texts parameterise by  $\beta = 1/\lambda$ , giving  $f_X(x) = \beta e^{-\beta x}$  with mean  $1/\beta$ .)

## Example 2.2.3 — Binomial Mean

### Example 2.2.3 (CB): Binomial( $n, p$ )

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n.$$

$$EX = \sum_{x=0}^n x \binom{n}{x} p^x (1 - p)^{n-x} = \sum_{x=1}^n x \binom{n}{x} p^x (1 - p)^{n-x}$$

Use the identity  $x \binom{n}{x} = n \binom{n-1}{x-1}$ :

$$EX = \sum_{x=1}^n n \binom{n-1}{x-1} p^x (1 - p)^{n-x}$$

$$= np \sum_{y=0}^{n-1} \binom{n-1}{y} p^y (1 - p)^{n-1-y} \quad (\text{set } y = x - 1)$$

$$= np \cdot 1 = \mathbf{np} \quad (\text{The sum is the total of a Binomial}(n-1, p) \text{ pmf} = 1.)$$

## Example 2.2.4 — Cauchy Mean Does Not Exist

### Example 2.2.4 (CB): Cauchy Distribution

$$f_X(x) = \frac{1}{\pi} \frac{1}{1+x^2}, \quad -\infty < x < \infty.$$

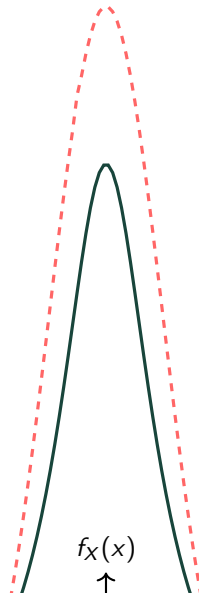
Does  $EX$  exist? Check  $E|X|$ :

$$E|X| = \int_{-\infty}^{\infty} |x| \cdot \frac{1}{\pi(1+x^2)} dx = \frac{2}{\pi} \int_0^{\infty} \frac{x}{1+x^2} dx = \frac{1}{\pi} \lim_{M \rightarrow \infty} \log(1+M^2) = \infty$$

### Conclusion

$E|X| = \infty$ , so  $EX$  **does not exist**.

The Cauchy distribution is the classic example of a distribution with *no mean*. Its heavy tails prevent the integral from converging.



# Linearity: $E(aX + b) = aEX + b$

## Linearity of Expectation (CB Eq. 2.2.1)

For any constants  $a$  and  $b$ :

$$E(aX + b) = aEX + b$$

**Proof (continuous case):**

$$\begin{aligned} E(aX + b) &= \int_{-\infty}^{\infty} (ax + b)f_X(x) dx \\ &= a \int x f_X(x) dx + b \int f_X(x) dx \\ &= aEX + b \cdot 1 = aEX + b. \quad \square \end{aligned}$$

## Applications

(i) If  $X \sim \text{Binomial}(n, p)$  with  $EX = np$ :

$$E(X - np) = np - np = 0$$

Centring at the mean gives zero expectation.

(ii) Temperature conversion: if  $C \sim \text{some distribution}$  with  $EC = 20^\circ$ :

$$EF = E\left(\frac{9}{5}C + 32\right) = \frac{9}{5}(20) + 32 = 68^\circ$$

## Theorem 2.2.5 — Properties of Expectation

### Theorem 2.2.5 (CB)

Let  $X$  be a random variable and  $a, b, c$  constants. For any functions  $g_1(x)$  and  $g_2(x)$  whose expectations exist:

- Ⓐ  $E(ag_1(X) + bg_2(X) + c) = aEg_1(X) + bEg_2(X) + c$  (**Linearity**)
- Ⓑ If  $g_1(x) \geq 0$  for all  $x$ , then  $Eg_1(X) \geq 0$  (**Nonnegativity**)
- Ⓒ If  $g_1(x) \geq g_2(x)$  for all  $x$ , then  $Eg_1(X) \geq Eg_2(X)$  (**Monotonicity**)
- Ⓓ If  $a \leq g_1(x) \leq b$  for all  $x$ , then  $a \leq Eg_1(X) \leq b$  (**Boundedness**)

### Why these matter

- (a) allows us to compute  $EX^2 = E[(X - \mu)^2] + (EX)^2$  and related identities without re-integrating.
- (b) and (c) give intuitive ordering: if  $X$  is always positive, so is its mean.
- (d) bounds the mean between the range of the function.

## Example 2.2.6 — EX Minimises Mean Squared Error

### Example 2.2.6 (CB): Minimising Distance

We want the constant  $b$  that best predicts  $X$  in the mean-squared sense. Minimise  $E(X - b)^2$  over  $b$ .

Add and subtract  $\mu = EX$ :

$$\begin{aligned} E(X - b)^2 &= E[(X - \mu) + (\mu - b)]^2 \\ &= E(X - \mu)^2 + 2(\mu - b) \underbrace{E(X - \mu)}_{=0} + (\mu - b)^2 \\ &= E(X - \mu)^2 + (\mu - b)^2 \end{aligned}$$

The first term is fixed. The second term  $(\mu - b)^2 \geq 0$  is minimised at  $b = \mu$ .

$$\min_b E(X - b)^2 = E(X - \mu)^2$$

### Interpretation

$EX$  is the **best constant predictor** of  $X$  under squared-error loss. This is the fundamental justification for using the mean as a measure of center.

# Two Methods for $Eg(X)$

## Two equivalent approaches

**Method 1 (Direct):** Use  $f_X$  directly:

$$Eg(X) = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

**Method 2 (Transform):** Find  $f_Y$  where  $Y = g(X)$ , then:

$$Eg(X) = EY = \int_{-\infty}^{\infty} y f_Y(y) dy$$

Both methods always give the same answer.

### Example 2.2.7 (CB): Uniform-Exponential II

$X \sim \text{Uniform}(0, 1)$ ,  $g(X) = -\log X$ .

**Method 1:**  $E(-\log X) = \int_0^1 (-\log x) \cdot 1 \, dx = [x - x \log x]_0^1 = 1$ .

**Method 2:** From Ex 2.1.4,  $Y = -\log X \sim \text{Exp}(1)$ , so  $EY = 1$  (by Ex 2.2.2 with  $\lambda = 1$ ).

Both give  $E(-\log X) = 1$ . ✓

## Additional Example A — Uniform Mean and $EX^2$

**Example A:**  $X \sim \text{Uniform}(a, b)$

$$f_X(x) = \frac{1}{b-a}, \quad a < x < b.$$

**Mean:**

$$EX = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \cdot \frac{b^2 - a^2}{2} = \frac{a+b}{2}$$

The mean is the *midpoint* of the interval — as expected by symmetry.

**Second moment:**

$$EX^2 = \int_a^b x^2 \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \cdot \frac{b^3 - a^3}{3} = \frac{a^2 + ab + b^2}{3}$$

(Use  $b^3 - a^3 = (b-a)(b^2 + ab + a^2)$ .)

## Additional Example B — Geometric Mean

**Example B:**  $X \sim \text{Geometric}(p)$

$$P(X = x) = (1 - p)^{x-1}p, \quad x = 1, 2, 3, \dots$$

$$EX = \sum_{x=1}^{\infty} x(1 - p)^{x-1}p = p \sum_{x=1}^{\infty} x(1 - p)^{x-1}$$

Use the identity  $\sum_{x=1}^{\infty} xr^{x-1} = \frac{1}{(1 - r)^2}$  for  $|r| < 1$ :

$$EX = p \cdot \frac{1}{(1 - (1 - p))^2} = p \cdot \frac{1}{p^2} = \frac{1}{p}$$

## Interpretation

If each trial succeeds with probability  $p$ , on average you need  $1/p$  trials to get the first success. E.g. fair coin ( $p = 1/2$ ): expect 2 tosses to get the first head. Rare event ( $p = 0.01$ ): expect 100 trials.

## Additional Example C — Jensen's Inequality (Preview)

### Jensen's Inequality (CB §2.2)

If  $g$  is a **convex** function ( $g'' \geq 0$ ), then

$$Eg(X) \geq g(EX)$$

If  $g$  is **concave** ( $g'' \leq 0$ ), the inequality reverses.

## Applications

(a)  $g(x) = x^2$  (convex):  $EX^2 \geq (EX)^2$ .

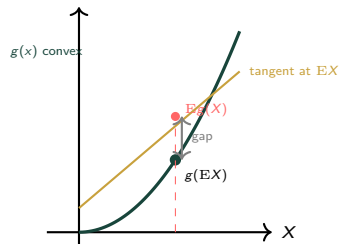
Rearranging:  $\text{Var}(X) = EX^2 - (EX)^2 \geq 0$ . ✓

(b)  $g(x) = \log x$  (concave,  $x > 0$ ):  $E \log X \leq \log EX$ .

(Arithmetic mean  $\geq$  geometric mean.)

(c)  $g(x) = e^x$  (convex):  $Ee^X \geq e^{EX}$ .

Used in moment generating functions.



## In-Class Activity — Section 2.2

Let  $X$  have pdf  $f_X(x) = 3x^2$  on  $[0, 1]$ , zero elsewhere.

**(a)** Find  $EX$ . **(b)** Find  $EX^2$ . **(c)** Find  $E(3X - 1)$  using linearity (Thm 2.2.5a).

$X$  has pmf  $P(X = -1) = \frac{1}{4}$ ,  $P(X = 0) = \frac{1}{2}$ ,  $P(X = 2) = \frac{1}{4}$ .

**(a)** Find  $EX$ . **(b)** Find  $EX^2$ . **(c)** Find the constant  $b$  that minimises  $E(X - b)^2$ . Verify using Ex 2.2.6.

Let  $X \sim \text{Uniform}(0, 1)$  and  $g(X) = X^2$ .

**(a)** Compute  $Eg(X)$  directly using  $f_X$ .

**(b)** Find  $f_Y(y)$  where  $Y = X^2$  (from §2.1), then compute  $EY = \int y f_Y(y) dy$ .

**(c)** Confirm both methods agree.

## Section 2.2 — Summary

| Concept               | Formula                         | Key result                    |
|-----------------------|---------------------------------|-------------------------------|
| Definition            | $Eg(X) = \int g(x)f_X(x)dx$     | Exists iff $E g(X)  < \infty$ |
| Linearity             | $E(aX + b) = aEX + b$           | Thm 2.2.5(a)                  |
| Exp( $\lambda$ ) mean | $EX = \lambda$                  | Ex 2.2.2                      |
| Binomial mean         | $EX = np$                       | Ex 2.2.3                      |
| Cauchy mean           | does not exist                  | Ex 2.2.4                      |
| Best predictor        | $\min_b E(X - b)^2$ at $b = EX$ | Ex 2.2.6                      |
| Jensen                | convex $g$ : $Eg(X) \geq g(EX)$ | §2.2                          |

## Two methods for $Eg(X)$

**Direct:**  $\int g(x)f_X(x) dx$  vs. **Transform:** find  $f_Y$  then  $\int y f_Y(y) dy$ .

Both give the same answer. Choose whichever integral is easier.

## Activity Solutions — Q1 & Q2

(a)  $EX = \int_0^1 x \cdot 3x^2 dx = 3 \int_0^1 x^3 dx = 3 \cdot \frac{1}{4} = \frac{3}{4}$

(b)  $EX^2 = \int_0^1 x^2 \cdot 3x^2 dx = 3 \int_0^1 x^4 dx = 3 \cdot \frac{1}{5} = \frac{3}{5}$

(c) By linearity:  $E(3X - 1) = 3EX - 1 = 3 \cdot \frac{3}{4} - 1 = \frac{9}{4} - 1 = \frac{5}{4}$

(a)  $EX = (-1) \cdot \frac{1}{4} + 0 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = -\frac{1}{4} + 0 + \frac{2}{4} = \frac{1}{4}$

(b)  $EX^2 = (-1)^2 \cdot \frac{1}{4} + 0^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{4} = \frac{1}{4} + 0 + \frac{4}{4} = \frac{5}{4}$

(c) By Example 2.2.6,  $E(X - b)^2$  is minimised at  $b = EX = \frac{1}{4}$ . Minimum value =  $E(X - \frac{1}{4})^2 = EX^2 - (EX)^2 = \frac{5}{4} - \frac{1}{16} = \frac{19}{16}$ .

## Activity Solutions — Q3 & Q4

(a) Direct:  $EX^2 = \int_0^1 x^2 \cdot 1 \, dx = \frac{1}{3}.$

(b) From §2.1 Ex 2.1.7:  $f_Y(y) = \frac{1}{2\sqrt{y}}, 0 < y < 1.$

$$EY = \int_0^1 y \cdot \frac{1}{2\sqrt{y}} \, dy = \int_0^1 \frac{\sqrt{y}}{2} \, dy = \frac{1}{2} \cdot \frac{2}{3} y^{3/2} \Big|_0^1 = \frac{1}{3}$$

(c) Both methods give  $EX^2 = \frac{1}{3}.$  ✓ This confirms the **Law of the Unconscious Statistician**.

(a)  $EX = \frac{1}{2}.$   $Ee^X = \int_0^1 e^x \, dx = [e^x]_0^1 = e - 1 \approx 1.718.$

(b)  $e^{EX} = e^{1/2} = \sqrt{e} \approx 1.649.$

(c)  $Ee^X = e - 1 \approx 1.718 > \sqrt{e} \approx 1.649 = e^{EX}.$  ✓

$g(x) = e^x$  is convex since  $g''(x) = e^x > 0$ , so Jensen gives  $Ee^X \geq e^{EX}.$